

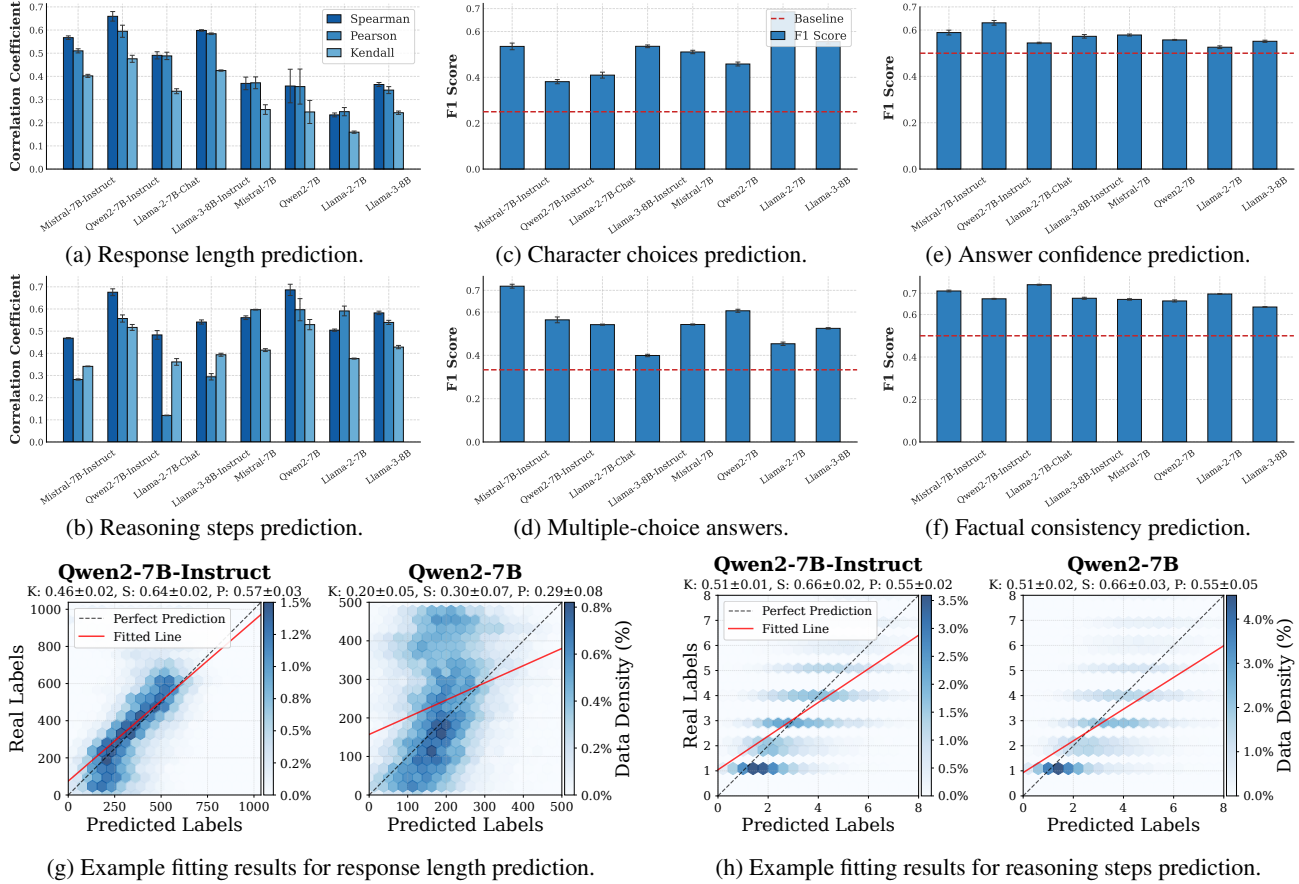


Figure 3. Cross-dataset generalization results. For regression tasks (response length, reasoning steps), correlations with targets remain strong despite reduced accuracy compared to in-dataset testing, as shown by Kendall (K), Spearman (S), and Pearson (P) coefficients. Classification tasks (character choices, multiple-choice, confidence, factual consistency) maintain above-baseline F1 scores. These results suggest the probes detect generalizable patterns rather than dataset-specific features, indicating transferable emergent planning capabilities within the task domain.

This creates a dataset of prompt representations and their future response attributes: $\mathcal{D} = \{\mathcal{H}_i, \hat{g}_i\}_{i=1}^N$. With this dataset, we then train a probe to predict targets from representations.

See Appendix A.3 for details on data collection, including task-specific and model-specific prompt templates, as well as data filtering and augmentation methods.

3.3. Experimental Details

Probe training. We train one-hidden-layer MLPs with ReLU activation, with hidden sizes chosen among $\mathcal{W} = \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024\}$. The output size is 1 for regression and the number of classes with a softmax layer for classification. Each probe is trained for 400 epochs using MSELoss for regression and CrossEntropyLoss for classification. Datasets are split 60 : 20 : 20 for train-validation-test. We perform a grid search over MLP hidden sizes $\mathcal{W}$ and representation layers $\mathcal{H}$ (as inputs to the probes), reporting the test scores for the best hyperpa-

rameters. Results are averaged across three random seeds.

Probe evaluation. For regression tasks (response length and reasoning steps), we evaluate using Spearman, Kendall and Pearson correlation coefficients, which measure the strength and direction of monotonic (Spearman, Kendall) and linear (Pearson) relationships between predicted and target values. For classification tasks, we evaluate using F1 scores: 4-class classification for character choices, 5-class classification for multiple-choice answers, and binary classification for answer confidence and factual consistency. In our setup, accuracy aligns with F1 score for classification due to strict class balance across tasks.

Language models. We experiment with both instruction-tuned models (Llama-2-7B-Chat, Llama-3-8B-Instruct, Mistral-7B-Instruct, and Qwen2-7B-Instruct) and their corresponding base models (Llama-2-7B, Llama-3-8B, Mistral-7B, and Qwen2-7B). See Appendix A.1 for model details.

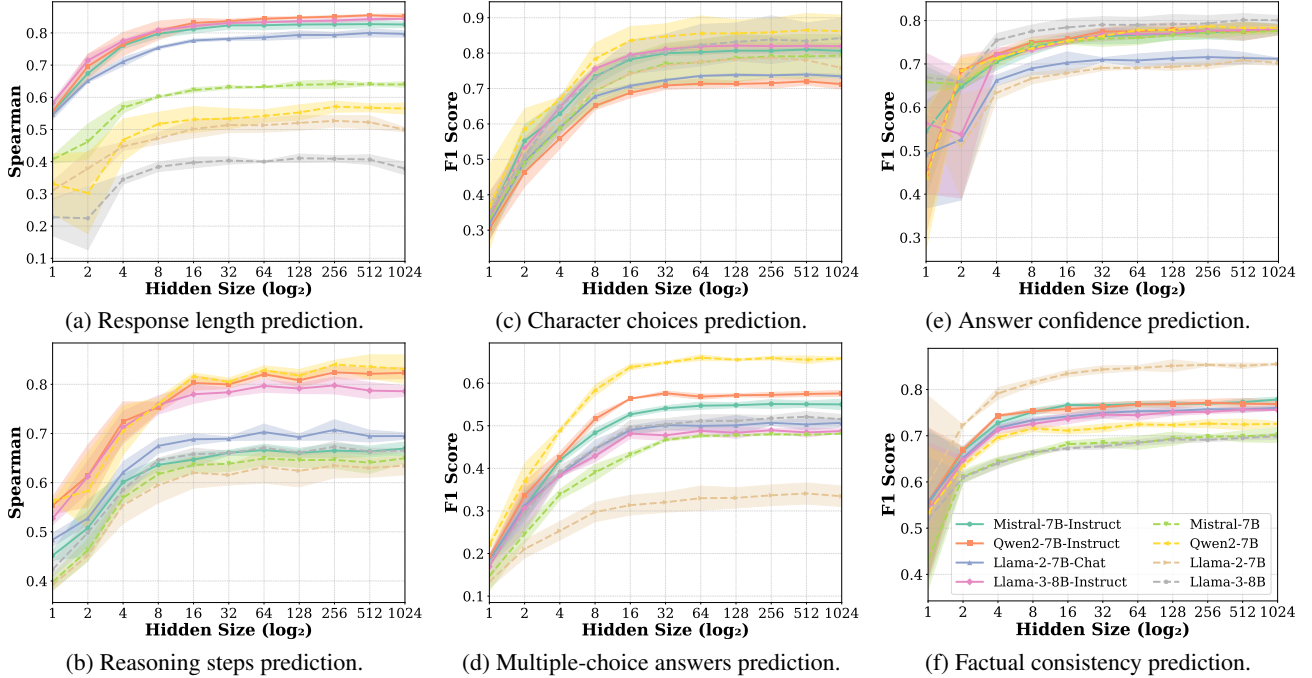


Figure 4. Hidden-size study results. Performance of MLP probes plateaus at relatively small hidden sizes (≤ 128) across all tasks, with structure attributes converging around size 16, content attributes at 32, and behavior attributes at 8. This suggests a hierarchy of pattern complexity, with behavioral patterns being most accessible and content patterns requiring more sophisticated probes.

4. Experimental Results

In this section, we present experimental results across six tasks, showing that LLM hidden prompt representations encode rich information about upcoming responses and can be used to probe and predict global response attributes.

Insight 1: Models present emergent planning on structure, content, and behavior attributes, which can be probed with high accuracy (Fig. 2). Our in-dataset probing experiments (where probes are trained and tested on different splits of the same prompt dataset) reveal that both base and fine-tuned models encode structure, content, and behavior attributes, with fine-tuned models showing superior performance. For structural attributes (response length and reasoning steps; Fig. 2a, 2b), fine-tuned models exhibit strong linear correlations with ground truth, clustering around $y = x$ (with example fitting results shown in Fig. 2g, 2h), while base models show weaker but positive correlations. For content and behavior attributes (character choices, multiple-choice answers, answer confidence, and factual consistency; Fig. 2c, 2d, 2e, 2f), both model types demonstrate robust classification performance above random baselines. These findings also suggest that **models develop systematic internal planning representations for content and behavior attributes during pre-training, with structure attributes requiring additional reinforcement through fine-tuning.**

Insight 2: The learned patterns generalize across datasets, indicating intrinsic task-related patterns rather than dataset-specific ones (Fig. 3). Our cross-dataset experiments (training and testing probes on different prompt datasets for the same task, e.g., GSM8K $\rightarrow$ MATH or TinyStories $\rightarrow$ ROCStories) demonstrate robust generalization of learned patterns. For structure attributes (Fig. 3a, 3b), predictions maintain strong correlations with target labels despite lower accuracy compared to in-dataset testing (with example fitting results shown in Fig. 3g, 3h), with fine-tuned models showing stronger correlations than base models. Similarly, for content and behavior attributes (Fig. 3c, 3d, 3e, 3f), performance remains above baseline in cross-dataset settings. These results suggest that **probes capture generalizable task-related patterns rather than dataset-specific features, indicating that models may develop intrinsic emergent planning capabilities that transfer across different contexts within the same task domain.**

Insight 3: Emergent planning patterns are salient across models and tasks, extractable with simple MLP probes (Fig. 4). We investigate pattern saliency by varying the hidden size of two-layer MLP probes and measuring their average performance across model layers. Performance plateaus before hidden size 128 across all datasets, with larger sizes that can even lead to overfitting, indicating pattern saliency. The results can also indicate saliency differences across attributes: structure attributes (Fig. 4a, 4b)

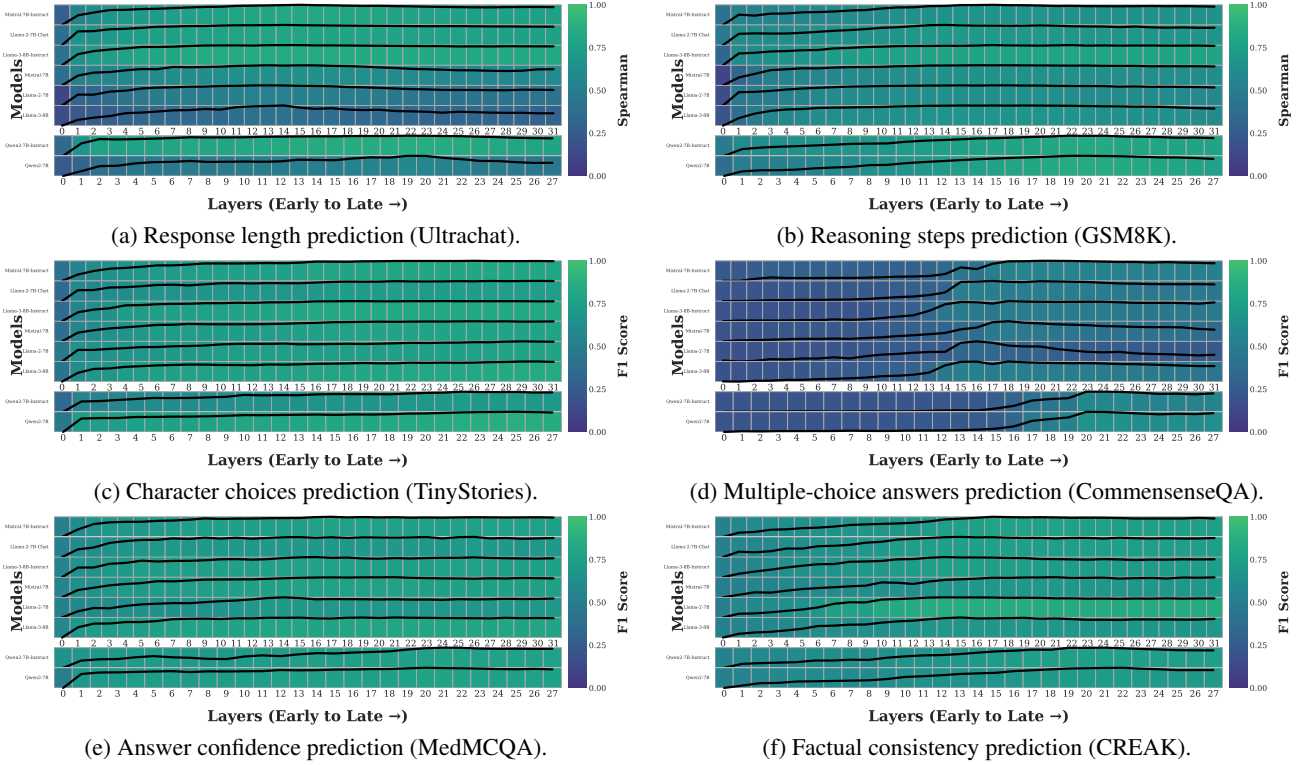


Figure 5. Layer-wise attribute prediction dynamics. Six subplots (one per task): Y-axis shows eight models; X-axis traces layer-wise progression (early $\rightarrow$ late). Heatmap colors indicate absolute performance (0–1, lighter = higher); black curves show row-normalized relative capability trends. Key dynamics: Structure attributes peak mid-layers, content attributes exhibit varied emergence timelines but consolidate in later layers, and behavior attributes stabilize early. Layer-wise probing reveals hierarchical organization of planning capabilities, with progressive refinement shaping final outputs.

converges around hidden size 16, content attributes (Fig. 4c, 4d) plateau around 32, and behavior attributes (Fig. 4e, 4f) plateaus at around 8, suggesting a hierarchy of representation complexity where behavioral patterns are most readily accessible, structural patterns require moderate complexity to capture, and content patterns demand the most sophisticated probe architectures. The consistent pattern across different model scales and architectures illustrates fundamental organizational principles in language model representations, suggesting that **emergent planning is an inherent property of large language models rather than an artifact of specific architectures or training procedures**.

Insight 4: Attribute patterns accumulate and peak differently across model layers (Fig. 5). We conduct layer-wise probing analysis (with hidden sizes optimized per layer) to understand how different attributes emerge through model layers. The results reveal distinct accumulation patterns for each attribute type. Structure attributes (Fig. 5a, 5b) show weak performance in early layers, peak in middle layers, and partially diminish in final layers, suggesting a gradual accumulation followed by refinement. Content attributes (Fig. 5c, 5d) peak in later layers, either through sudden

late-layer emergence or gradual accumulation, indicating their reliance on higher-level semantic processing. Behavior attributes (Fig. 5e, 5f) demonstrate uniform distribution across layers except for the initial few, suggesting they are fundamental properties encoded early in the model. These layer-wise patterns reveal that (1) different aspects of planning emerge through distinct computational paths, (2) the hierarchical nature of planning, from basic behavioral patterns to complex structural decisions, is reflected in the layer-wise organization, and (3) the emergence of these patterns through progressive transformations, rather than from initial embeddings alone, indicates that planning capabilities arise from learned computational processes rather than simple statistical correlations.

5. Ablation

5.1. Planning Ability Scales with Model Size

We analyze how emergent response planning scales across different model sizes using four model families: Llama-2-chat (7B, 13B, 70B), Llama-3-Instruct (8B, 70B), Qwen-2-Instruct (7B, 72B), and Qwen-2.5-Instruct (1.5B, 32B, 72B). Using grid search over layers and hidden sizes, we identify